

Anthony Bardou

RESEARCH SCIENTIST – MACHINE LEARNING, OPTIMIZATION & AI

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Research scientist specializing in machine learning and optimization for decision-making under uncertainty. My work combines theoretical analysis with practical algorithm design, with a focus on Bayesian optimization, online learning, and high-dimensional black-box problems in dynamic and decentralized settings. I also explore efficient evaluation methods for language models and develop open-source research software. My work has been published at ICLR, NeurIPS, and in leading machine-learning and networking journals, and has received best paper and best PhD thesis awards.

Core Expertise

ML & AI	Bayesian Optimization, Online Learning, Bandits, LLM Evaluation, Probabilistic Models
Optimization	Zeroth-Order, Stochastic, High-Dimensional, Dynamic & Decentralized Optimization
Engineering	Python, PyTorch, BoTorch, NumPy/SciPy, C/C++, Git, Linux
Languages	French (native), English (full professional proficiency), Spanish and Romanian (basic)

Experience

École Polytechnique Fédérale de Lausanne

Lausanne, CH

POSTDOCTORAL RESEARCHER – MACHINE LEARNING AND OPTIMIZATION

Oct. 2023 - Present

- Develop scalable Bayesian optimization methods for noisy, high-dimensional, dynamic, and symmetry-rich black-box problems, combining theoretical guarantees with empirical evaluation.
- Published at ICLR and NeurIPS; released reusable PyTorch/BoTorch implementations and contributed to a benchmark for diffusion language models.
- Collaborate with researchers across machine learning, optimization, and networked systems; supervised nine Bachelor's and Master's research projects.

Keywords: Bayesian Optimization, Online Learning, Dynamic Systems, PyTorch, BoTorch

École Normale Supérieure de Lyon

Lyon, FR

RESEARCH INTERN – ONLINE LEARNING FOR WIRELESS SYSTEMS

Apr. 2020 - Sep. 2020

- Designed multi-armed-bandit methods for adaptive spatial reuse in dense Wi-Fi networks and evaluated them in a custom network simulator.
- Established the research direction later developed into an award-winning PhD thesis and multiple peer-reviewed publications.

Keywords: Online Learning, Multi-Armed Bandit, Spatial Reuse

Université Claude Bernard Lyon I

Lyon, FR

RESEARCH INTERN – NATURAL LANGUAGE PROCESSING

Dec. 2018 - Jun. 2019

- Developed attention and representation-learning methods to classify citation intent in scientific documents, improving F1 by 23 points on internal benchmarks.

Keywords: Natural Language Processing, Embedding, Attention, Deep Learning, PyTorch

Université de Perpignan Via Domitia

Perpignan, FR

UNDERGRADUATE RESEARCH INTERN – OPTIMIZATION AND COMPUTER VISION

Apr. 2018 - Jun. 2018

- Analyzed the robustness of a non-invasive computer-vision system for respiratory-volume measurement and reduced its computational complexity using combinatorial optimization.

Keywords: Combinatorial Optimization, Robustness Analysis, Image Processing

Education

2020–23	PhD in Machine Learning and Wireless Networks , École Normale Supérieure de Lyon — <i>Online Learning for the Black-Box Optimization of Wireless Networks</i> ; Best PhD Thesis Award	Lyon, FR
2018–20	M.S. in Data Science , Université Claude Bernard Lyon I — ranked 1st of 47, highest honors	Lyon, FR
2015–18	B.S. in Computer Science and Mathematics , Université de Perpignan Via Domitia — ranked 1st of 16, highest honors	Perpignan, FR

Honors & Awards

INTERNATIONAL

2025	Top Reviewer , 39th Annual Conference on Neural Information Processing Systems	San Diego, US
2024	Top Reviewer , 38th Annual Conference on Neural Information Processing Systems	Vancouver, CA
2022	Best Paper Award , 25th International Conference on Modeling, Analysis and Simulation of Wireless and Mobile Systems	Montreal, CA

NATIONAL

2025	Best Ph.D. Thesis , GDR Réseaux et Systèmes Distribués / Association ACM SIGOPS France	Toulouse, FR
2022	Best Student Paper Award , 24ème Rencontres Francophones sur les Aspects Algorithmiques des Télécommunications	Paris, FR

Selected Open-Source Research Software

2026	speedrun-dlm , 170M-parameter diffusion-LM speedrun on 8 H100s; seven objectives, 50 seeds, 6,400 samples, quality and inference-FLOP constraints	PyTorch
2026	max-kernel , Drop-in symmetry-aware GP kernel using PSD projection and Nyström extension; published at ICLR	BoTorch
2024	W-DBO , Pip-installable dynamic optimizer with online stale-data removal and a performance-critical C++ criterion; published at NeurIPS	Python/C++
2023	DuMBO , Parallel message-passing optimizer for high-dimensional black boxes with an ask/tell API; published at ICLR	BoTorch

Talks

Jul. 2026	Invited Speaker , L2S (Paris Saclay University, CentraleSupélec)	Paris, FR
Sep. 2025	Invited Speaker , ICube Lab (Strasbourg University)	Strasbourg, FR
May 2025	Invited Speaker , Secondmind Research Seminar	Cambridge, UK
Mar. 2025	Invited Speaker , Journées GDR Réseaux et Systèmes Distribués	Toulouse, FR
Jun. 2024	Invited Speaker , INRIA Maracas (INSA Lyon)	Lyon, FR
May 2023	Invited Speaker , INDY Lab (EPFL)	Lausanne, CH
Oct. 2022	Oral Presentation , MSWiM'22	Montreal, CA
Jul. 2022	Invited Speaker , INRIA Maracas (INSA Lyon)	Lyon, FR
Jun. 2022	Oral Presentation , AlgoTel'22	Paris, FR
Feb. 2022	Oral Presentation , ROADEF'22	Lyon, FR
Dec. 2021	Invited Speaker , INRIA Ockham (ENS Lyon)	Lyon, FR
May 2021	Invited Speaker , INRIA Maracas (INSA Lyon)	Lyon, FR
Nov. 2020	Invited Speaker , Performance Evaluation for Networks	Lyon, FR
Jul. 2018	Invited Speaker , LIRMM (Perpignan University)	Perpignan, FR

Schools & Workshops

Swiss Optimization Symposium (SOS'26)	Aug. 2026	Ascona, CH
Spring School of Theoretical Computer Science on Machine Learning (EPIT'22)	May 2022	Luminy, FR

External Accolades

Prix de thèse du GDR RSD : des travaux sur l'optimisation récompensés	Jan. 2025	INS2I (CNRS)
Lauréat du prix de thèse GDR RSD - Édition 2024	Sep. 2024	GDR RSD (CNRS)
L'optimisation bayésienne au service des performances du WiFi	Apr. 2023	INS2I (CNRS)

Leadership, Teaching & Service

PEER REVIEW (YEAR FIRST REVIEWED)

2025	International Conference on Learning Representations , International Conference
2025	Scientific Reports , Peer-Reviewed Journal (Nature)
2025	Transactions on Machine Learning Research , Peer-Reviewed Journal
2025	International Conference on Machine Learning , International Conference
2024	Computer Networks , Peer-Reviewed Journal (Elsevier)
2023	Neural Information Processing Systems , International Conference
2023	Computer Communications , Peer-Reviewed Journal (Elsevier)
2023	International Symposium on Information Theory , International Conference (IEEE)

TEACHING

Fall 2024–26	Stochastic Modeling , 30 hours @ EPFL (Bachelor)	Lausanne, CH
Spring 2023	Preparation to the <i>agrégation</i> of Computer Science , 32 hours @ ENS Lyon (Master)	Lyon, FR
Spring 2022	Numerical Methods , 15 hours @ UCBL (Bachelor)	Lyon, FR
Spring 2022	Optimization , 15 hours @ UCBL (Bachelor)	Lyon, FR
Fall 2021	Introduction to Artificial Intelligence , 12 hours @ UCBL (Master)	Lyon, FR
Fall 2021	Optimization and Operations Research , 18 hours @ UCBL (Master)	Lyon, FR

SUPERVISION AND MENTORING

Spring 2026	Marin Philippe , Master Thesis @ EPFL	Lausanne, CH
Spring 2025	Robin Jaccard , Master Thesis @ EPFL	Lausanne, CH
Spring 2025	Adrien Deschenaux , Bachelor Semester Project @ EPFL	Lausanne, CH
Spring 2025	Aryan Ahadinia , Master Semester Project @ EPFL	Lausanne, CH
Fall 2024	Antoine Merel , Master Semester Project @ EPFL	Lausanne, CH
Fall 2024	Nahuel Plomb Veuve , Bachelor Semester Project @ EPFL	Lausanne, CH
Spring 2024	Emanuele Mengoli , Master Thesis @ EPFL (from École Polytechnique)	Lausanne, CH
Spring 2024	Giovanni Ranieri , Bachelor Semester Project @ EPFL	Lausanne, CH
Spring 2023	Hugo Martel , Bachelor Research Internship @ ENS Lyon	Lyon, FR

JURIES AND COMMITTEES

Jul. 2026	External Expert , Master Thesis Defense of David Kalajdzic	EDIC, EPFL (CH)
Dec. 2025	Committee Member , Evaluation of prospective PhD students	EDIC, EPFL (CH)
May 2024	External Expert , Evaluation of prospective PhD students	INRIA Lyon (FR)

Publications (11 peer-reviewed papers)

INTERNATIONAL CONFERENCES

ANTHONY BARDOU, ANTOINE GONON, ARYAN AHADINIA, PATRICK THIRAN. Symmetry-Aware Bayesian Optimization via Max Kernels. *Proc. ICLR'26*. *Rio de Janeiro, BR*. Apr. 2026

ANTHONY BARDOU, PATRICK THIRAN, GIOVANNI RANIERI. This Too Shall Pass: Removing Stale Observations in Dynamic Bayesian Optimization. *Proc. NeurIPS'24*. *Vancouver, CA*. Dec. 2024

ANTHONY BARDOU, PATRICK THIRAN, THOMAS BEGIN. Relaxing the Additivity Constraints in Decentralized No-Regret High-Dimensional Bayesian Optimization. *Proc. ICLR'24*. *Vienna, AT*. May 2024.

ANTHONY BARDOU, THOMAS BEGIN. INSPIRE: Distributed Bayesian Optimization for ImproviNg SPatIal REuse in Dense WLANs. *Proc. MSWiM'22*. *Montréal, CA*. Oct. 2022. 10.1145/3551659.3559050. **BEST PAPER AWARD**.

ANTHONY BARDOU, THOMAS BEGIN, ANTHONY BUSSON. Improving the Spatial Reuse in IEEE 802.11ax WLANs: A Multi-Armed Bandit Approach. *Proc. MSWiM'21*. *Alicante, ES*. Nov. 2021. 10.1145/3479239.3485715.

PEER-REVIEWED JOURNALS

ANTHONY BARDOU, JEAN-MARIE GORCE, THOMAS BEGIN. Assessing the Performance of NOMA in a Multi-Cell Context: A General Evaluation Framework. *IEEE Transactions on Wireless Communications*. Jul. 2025.

SAMIR SI-MOHAMMED, **ANTHONY BARDOU**, THOMAS BEGIN, ISABELLE GUÉRIN-LASSOUS, PASCALE VICAT-BLANC. NS+ NDT: Smart Integration of Network Simulation in Network Digital Twin, Application to IoT Networks. *Future Generation Computer Systems*. Mar. 2024.

ANTHONY BARDOU, THOMAS BEGIN. Analysis of a Decentralized Bayesian Optimization Algorithm for Improving Spatial Reuse in Dense WLANs. *Computer Communications, ACM MSWiM 2022 Special Issue*. Jun. 2023. 10.1016/j.comcom.2023.06.004

ANTHONY BARDOU, THOMAS BEGIN, ANTHONY BUSSON. Mitigating Starvation in Dense WLANs: A Multi-Armed Bandit Solution. *Ad Hoc Networks*, Vol. 138. Jan. 2023. 10.1016/j.adhoc.2022.103015

ANTHONY BARDOU, THOMAS BEGIN, ANTHONY BUSSON. Analysis of a Multi-Armed Bandit Solution for Improving the Spatial Reuse of New Generation WLANs. *Computer Communications, ACM MSWiM 2021 Special Issue*. Sep. 2022. 10.1016/j.comcom.2022.07.015

INTERNATIONAL WORKSHOPS

ANTOINE GONON, ADRIAN MÜLLER, LÉON ZHENG, CLÉMENT LALANNE, ZEBANG SHEN, YA-PING HSIEH, **ANTHONY BARDOU**, NICOLAS BOUMAL. Measuring Progress in Diffusion Language Model Pretraining. *NeurIPS'26 Workshop BeNTe*. Sydney, AU. Dec. 2026

PH.D. THESIS

ANTHONY BARDOU. Online Learning for the Black-Box Optimization of Wireless Networks. ENS Lyon. Lyon, FR. Sep. 2023. **GDR RSD/ASF BEST PH.D. THESIS AWARD.**